

Hyperbolic Flavor Geometry — SSRN Preprint Portfolio

In Support of Account Reinstatement Appeal
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Program Overview

The Hyperbolic Flavor Geometry (HFG) program establishes that the Standard Model flavor parameters — fermion masses, CKM and PMNS mixing matrices, CP violation — arise from the arithmetic topology of two compact hyperbolic 3-manifolds, M_{PMNS} and M_{CKM} . All 15 preprints present rigorous mathematical results with computational verification using SnapPy, a standard software package for hyperbolic 3-manifold computation developed at the University of Illinois and Monash University. Active journal submissions are listed for each paper.

Preprints (in order of SSRN Abstract ID)

1. SSRN 6583549 — *Distributed (SSRN-approved)*

Twist Angle Spectrum of Hyperbolic Holonomy

Under review: Results in Physics (RINP-D-26-00330)

Foundational paper. Establishes that loxodromic twist angles of $\pi_1(M)$ encode Standard Model flavor parameters. Proves the A-factor decomposition theorem.

2. SSRN 6583550 — *Distributed (SSRN-approved)*

Quark Mixing from Hyperbolic Geometry: CKM Matrix

Under review: Results in Physics (RINP-D-26-00327)

Derives the CKM quark mixing matrix from holonomy of $m006(-5, 2)$. Optimal fitness 0.01729 vs PDG values.

3. SSRN 6583551 — *Distributed (SSRN-approved)*

CP Violation from A-Factor Twist Angles

Under review: Results in Physics (RINP-D-26-00329)

Derives the Jarlskog CP-violation invariant geometrically. Proves chirality-CP theorem.

4. SSRN 6583553 — *Distributed (SSRN-approved)*

Lepton Mixing from Borel Structure of Hyperbolic Holonomy

Under review: Results in Physics (RINP-D-26-00328)

Derives the PMNS lepton mixing matrix from holonomy of $m003(-2, 3)$. Optimal fitness 0.01379 vs PDG values.

5. SSRN 6631218 — *THE FLAGGED PAPER*

Neutrino Masses from the Golden Ratio Lattice: A Parameter-Free Prediction with Arithmetic Uniqueness

Under review: Journal of Physics G (JPhysG-105833)

This paper presents proved mathematical theorems about the golden ratio mass lattice $m = m_e \cdot \varphi^{q/4}$. The phrase “parameter-free” refers to the mathematical construction having *no free parameters* — all mass ratios follow from the arithmetic of $\varphi = (1 + \sqrt{5})/2$ without tuning. This is a theorem, not a speculative claim. The phrase “arithmetic uniqueness” refers to the proved result that the quarter-integer indices consistent with the three-generation constraint are unique. The paper has been submitted to and is under peer review at the Journal of Physics G, a refereed journal of the Institute of Physics.

6. SSRN 6660598 — *In Review at SSRN*

The Lucas Sequence as the Arithmetic Bridge in HFG

Submitted: Journal of Number Theory (JNTH-D-26-00538)

Proves covering tower primes of M_{PMNS} are exactly the prime Lucas numbers $\{2, 3, 7, 11, 29\}$. Verified across 11,031 manifolds.

7. SSRN 6669600

Golden Ratio Quantisation of Gravitational Wave Peak Frequencies

Preprint only. Connects HFG to Maleknejad-Kopp PRL 2026.

Falsifiable predictions for Einstein Telescope and Cosmic Explorer.

8. SSRN 6670398

HFG Conjectures on Dark Sector, Dark Energy, and Information Geometry

Preprint only.

Three precisely formulated, falsifiable conjectures extending HFG beyond the Standard Model flavor sector.

9. SSRN 6670778

Universal Spectral Phase Transition in Dehn Fillings of $m003$ and $m006$

Preprint only (short version).

Proves the three-generation threshold theorem: $1/6 < (\log \varphi)^2 < 1/3$. Verified by twelve independent methods from seven mathematical disciplines.

10. SSRN 6687838

Topologically Protected Qubit Gate Configurations

Submitted: Journal of Mathematical Physics (JMP26-AR-01272)

Proves Covering Tower Invariance and Homology Class No-Go theorems connecting hyperbolic topology to quantum information theory.

11. SSRN 6688021

Homology Class Asymmetry in the Loxodromic Twist Spectrum

Resubmitting to a topology journal.

Census of 1,062,802 loxodromic words. Introduces the spectral floor invariant and proves a paired hierarchy of decay rates.

12. SSRN 6689419

The Weeks Manifold, Dehn Surgery, and the Lepton Sector

Resubmitting to Journal of Geometry and Physics.

Proves CS invariant preservation $CS = 1/4$ under Dehn surgery. Semiclassical partition functions match volume ratios to 5–6 significant figures.

13. SSRN 6689441

Discrete Mixing Operators from Boundary Sector Geometry

Resubmitting to Communications in Contemporary Mathematics.

Three proved theorems: Geometric Dependence, Stability, and Discreteness of admissible unitary mixing matrices.

14. SSRN 6689678

Geometric Origin of CP Phases from Hyperbolic Holonomy

Resubmitting to Nuclear Physics B.

Proves CP violation arises as a topological invariant of flat $U(1)$ connections. Stable under all metric deformations.

15. SSRN 6689898

Arithmetic of Dehn Filling Slopes and the Flavor Covering Tower

Submitted: Michigan Mathematical Journal.

Falsifiable arithmetic fingerprint: covering tower primes = $\{2, 3, 5, 7, 11, 13, 29\}$, each with a precise source in the Dehn filling slopes. Zero violations across 24 covers through degree 6.

Summary

- All 15 papers present proved theorems and/or computational results with full reproducibility
- 4 papers already approved and distributed by SSRN (Abstract IDs 6583549, 6583550, 6583551, 6583553)
- Active peer review at 8 refereed journals
- Computational verification using SnapPy (standard software, NSF-funded, open source)
- ORCID: 0009-0006-4550-2663 (verified)
- No speculative content: all claims are proved theorems or falsifiable predictions

I respectfully request reinstatement of my account so that this research program can remain accessible to the scholarly community.